

Shijie Liu

DOCTORAL RESEARCHER · COMPUTATIONAL FLUID DYNAMICS

Chongqing University, Chongqing, China

☎ (+86) 151-3169-1003 | ✉ 20241601053@stu.cqu.edu.cn | 🏠 shijieliu.com | 📱 Jason-S-Liu | 🆔 0009-0000-5048-2536 | 🔍 Research: bluff-body aerodynamics, particle-laden flows, reduced-order modeling

Mechanism-driven CFD for particle-laden bluff-body wakes.

Education

Chongqing University

PH.D. STUDENT IN CIVIL ENGINEERING; ADVISOR: PROF. YUTONG LI

Chongqing, China

Sep. 2024 - Present

Shijiazhuang Tiedao University

M.ENG. IN CIVIL AND HYDRAULIC ENGINEERING; ADVISOR: PROF. QINGKUAN LIU

Shijiazhuang, China

Sep. 2021 - Jun. 2024

- Thesis: Data-driven unsteady aerodynamic modeling and reduced-order flow-field modeling for bluff-body structures.

Hebei University of Science and Technology

B.ENG. IN CIVIL ENGINEERING

Shijiazhuang, China

Sep. 2017 - Jun. 2021

Research Interests

Fluid Mechanics

Bluff-body aerodynamics, vortex shedding, wake transition, turbulent flow, wind engineering

Particle Flows

Inertial particle transport, clustering, void formation, deposition, Lagrangian particle tracking

Data-Driven Modeling

Dynamic mode decomposition, reduced-order modeling, LSTM-based aerodynamic prediction

Research Projects

Reduced-order Modeling of Unsteady Flow Fields and Aerodynamic Prediction Based on Machine Learning Graduate Innovation Project, Hebei Province

PRINCIPAL INVESTIGATOR

Jan. 2023 - Jan. 2024

- Developed an LSTM-based time-series model for unsteady aerodynamic force prediction.
- Built a robust reduced-order modeling workflow for unsteady flow-field reconstruction and analysis.

Intelligent Turbulence Modeling for Cavitating Flow

National Natural Science
Foundation of China Young
Scientists Fund

CORE MEMBER

Jan. 2023 - Dec. 2025

- Contributed to neural-network model development and data post-processing.

Key Technologies for Windbreak Nets at Outdoor Snow Venues of the Winter Olympics

Wind Tunnel Experiment

RESEARCH PARTICIPANT

Oct. 2021 - Dec. 2022

- Assisted with wind-tunnel model construction, pressure measurements, data acquisition, and data processing.

Wind-Resistance Study of the Lupar Cross-Sea Bridge in Malaysia

Wind Engineering

RESEARCH PARTICIPANT

Jan. 2021 - Feb. 2023

- Assisted with wind-tunnel model construction, vibration measurements, data acquisition, and data processing.

Skills

Programming

Python, TensorFlow, C++, LaTeX

CFD

OpenFOAM, turbulence modeling, Lagrangian particle tracking, benchmark-case setup

Pre/Post-processing

ICEM CFD, Tecplot, Origin, mesh generation, data visualization

Engineering

AutoCAD, boundary-layer wind-tunnel testing, pressure/force/vibration measurements

Languages

Chinese, English; CET-4: 532, CET-6: 513

Honors & Awards

2023-2024 **Top Ten Academic Stars of Graduate Students**, Shijiazhuang Tiedao University
2022-2023 **National Scholarship for Graduate Students**, Ministry of Education of China
2021-2022 **Excellent Graduate Student**, Shijiazhuang Tiedao University
2023 **Excellent Paper Award**, 32nd National Conference on Structural Engineering
2019-2020 **Third Prize**, Pinming Cup National BIM Competition
2018-2019 **First-Class Academic Scholarship**, Hebei University of Science and Technology

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Academic Activities

Forum for Young and Middle-Aged Experts in Building Structures

Xi'an, China

PRESENTER

2024

- Talk: Flow-field excitation source analysis based on Koopman LTI theory and data-driven algorithms.

13th National Conference on Fluid Mechanics

Harbin, China

PRESENTER

2024

- Talk: Flow-field excitation source analysis based on Koopman LTI theory and data-driven algorithms.

20th International Advanced School on Wind Engineering

Chongqing, China

STUDENT PARTICIPANT

2024

32nd National Conference on Structural Engineering

Jiangxi, China

PRESENTER

2023

- Talk: Unsteady flow-field analysis of flow around a square cylinder based on dynamic mode decomposition.

21st National Conference on Structural Wind Engineering

Hunan, China

STUDENT PARTICIPANT

2023

2nd Academic Conference on Extreme Loads and Protection of Bridge Structures

Shijiazhuang, China

CONFERENCE VOLUNTEER

May 2023

- Supported conference organization and reporting for an event with more than 400 participants from about 80 institutions.

Publications

Analysis and reconstruction of the wake flow on girder of a long-span bridge using Dynamic Mode Decomposition

Physics of Fluids

JOURNAL ARTICLE

2025

- **Liu Shi-jie**, Zhang Zhen, Qin Zejun, Zhang Yan, Jing Hongmiao, Liu Qing-kuan. SCI manuscript under submission.

Reconstruction and short-term prediction of wind pressure field on cylindrical coal sheds under undisturbed conditions using dynamic mode decomposition

Journal of Wind Engineering and Industrial Aerodynamics

JOURNAL ARTICLE

2025

- Liu Qing-kuan, Huo Jing, **Liu Shi-jie**, Qin Ze-jun, Li Hao-han, Zhang Zhen. 258: 106017.

LSTM-based aerodynamic modeling for unsteady flows around structures

Wind and Structures

JOURNAL ARTICLE

2024

- **Liu Shi-jie**, Zhang Zhen, Zhou Xue, Liu Qing-kuan. 38(2): 147-160.

Bridge unsteady aerodynamic force prediction based on long short-term memory networks

China Journal of Highway and Transport

JOURNAL ARTICLE

2023

- Liu Qing-kuan, **Liu Shi-jie**, Zhang Zhen, Zhou Xue, Jing Hongmiao. 36(8): 56-65.

Unsteady flow-field analysis of flow around a square cylinder based on dynamic mode decomposition

Engineering Mechanics

JOURNAL ARTICLE

2023

- **Liu Shi-jie**, Zhang Zhen, Liu Qing-kuan.

Selected papers in structural engineering and bluff-body flow modeling

National conferences

CONFERENCE PAPERS

2022-2023

- **Liu Shi-jie**, Zhang Zhen, Liu Qing-kuan. DMD-based unsteady flow-field analysis around a square cylinder.
- **Liu Shi-jie**, Hu Bo, Zhang Zhen, Liu Qing-kuan. LSTM-based aerodynamic coefficient prediction for two-dimensional cylinder flow.
- Huo Jing, **Liu Shi-jie**, Zhang Zhen, Liu Qing-kuan. Multi-scale LSTM prediction of wind pressure on coal-shed surfaces.